

# Oja-Type Plasticity and Neural Signal Reconstruction in Recurrent Networks

A Dynamical Systems Approach Using Bifurcation Analysis and Random Matrix Theory

David Hovey & Aditya Biswas

Princeton University

May 8, 2026

# Motivation: Brain Rhythms, Dimensionality, and Seizure-Like Synchrony

## Why study rhythmic neural activity?

- Neural circuits process time-varying signals: sensory inputs, motor rhythms, and internal oscillations.
- These signals can often be understood as combinations of simpler oscillatory components, like a Fourier series.
- Good representation requires rich, flexible activity rather than every neuron doing the same thing.

## What changes in seizure-like states?

- Seizures involve abnormal synchronization across neural populations.
- When many neurons fire together, the

### Central Question

Does Oja-type plasticity under rhythmic input preserve high-dimensional activity, or drive the network toward low-dimensional synchrony?

### Project idea:

We study whether a simple local learning rule can reshape the stability of a recurrent network and create a pathway toward synchronized dynamics.

# Background: Fixed-Weight Recurrent Networks

A common starting point is a recurrent neural network whose synaptic weights are fixed in time:

$$\tau \dot{x}_i = -x_i + f \left( \sum_j W_{ij} x_j + b_i \right).$$

Here  $x_i(t)$  is the activity of neuron  $i$ ,  $W_{ij}$  is the weight from neuron  $j$  to neuron  $i$ ,  $b_i$  is an external input or bias, and  $f$  is a nonlinear activation function.

In this setting, memory and computation are usually studied through the evolving neural state  $\mathbf{x}(t)$ , while the matrix  $W$  defines the fixed recurrent architecture.

## Dynamical systems viewpoint

For fixed  $W$ , stability, oscillations, and chaos are controlled by the interaction between the weight spectrum and the nonlinear activity dynamics.

# Background: The Grossberg $S$ - $\Sigma$ Viewpoint

A recurrent rate model is often written in the form

$$\tau \dot{x}_i = -x_i + f \left( \sum_j W_{ij} x_j + b_i \right).$$

In this formulation, the nonlinear activation is applied after neuron  $i$  integrates its total recurrent input.

An alternative formulation, common in Grossberg-type models, is

$$\tau \dot{x}_i = -x_i + \sum_j W_{ij} S(x_j) + b_i.$$

Here  $S(x_j)$  is interpreted as the nonlinear signal emitted by the presynaptic neuron  $j$ , which is then weighted by  $W_{ij}$  and summed by the postsynaptic neuron.

## Formal interpretation

The  $S$ - $\Sigma$  viewpoint separates the internal activity variable  $x_i$  from the transmitted

# Background: Learning in Recurrent Neural Networks

In the fixed-weight setting, the state  $\mathbf{x}(t)$  evolves through a fixed recurrent architecture  $W$ . Introducing learning makes the weights dynamic, so the system evolves in the enlarged state space

$$(\mathbf{x}(t), W(t)).$$

A standard local model of synaptic modification is Hebbian learning,

$$\dot{W}_{ij} = \eta x_i x_j,$$

which increases the weight  $W_{ij}$  when the presynaptic and postsynaptic activities are correlated. However, this rule alone can lead to unbounded growth of the weights.

Oja's rule modifies Hebbian learning by adding an activity-dependent normalization term:

$$\dot{W}_{ij} = \eta x_i x_j - \eta x_i^2 W_{ij}.$$

Oja-type plasticity preserves the local Hebbian dependence on neural activity while keeping the weights bounded. As a result, the recurrent architecture  $W(t)$  evolves with the neural state and can change the stability and long-term behavior of the

# Background: Girko's Circular Law

Let  $W^{(n)} = (W_{ij}) \in \mathbb{R}^{n \times n}$  be a random matrix whose entries are independent and identically distributed (i.i.d.) real random variables satisfying

$$\mathbb{E}[W_{ij}] = 0, \quad \mathbb{E}[W_{ij}^2] = \frac{g^2}{n}, \quad \mathbb{E}[|W_{ij}|^{2+\epsilon}] < \infty.$$

Let  $\lambda_1, \dots, \lambda_n \in \mathbb{C}$  denote the eigenvalues of  $W^{(n)}$ , and define the empirical spectral measure

$$\mu_n = \frac{1}{n} \sum_{k=1}^n \delta_{\lambda_k}.$$

Then Girko's circular law states that, as  $n \rightarrow \infty$ ,

$$\mu_n \Rightarrow \mu_{\text{circ}},$$

where  $\Rightarrow$  denotes weak convergence of probability measures and

$$d\mu_{\text{circ}}(z) = \frac{1}{\pi g^2} \mathbf{1}_{\{|z| \leq g\}} d^2 z$$

is the uniform probability measure on the disk of radius  $g$  in  $\mathbb{C}$ .

## Interpretation for recurrent networks

# Fixed-Point Analysis

For Adi To-do

# Special Case: Internally Driven Fixed-Weight Dynamics

First consider the case with no external drive and zero bias:

$$\tau \dot{x} = -x + \tanh(Wx).$$

All activity is generated internally by the recurrent network. Since  $\tanh(0) = 0$ , the origin is an equilibrium:

$$x^* = 0.$$

Linearizing at  $x = 0$ , using  $\tanh'(0) = 1$ , gives

$$J(0) = \frac{1}{\tau}(-I + W).$$

If  $W_{ij}$  are random with

$$\mathbb{E}[W_{ij}] = 0, \quad \text{Var}(W_{ij}) = \frac{g^2}{n},$$

then Girko's circular law implies that the eigenvalues of  $W$  approximately fill the disk  $|z| \leq g$ .



# Consequence: Local Stability Threshold

The linearized dynamics near  $x = 0$  are determined by

$$J(0) = \frac{1}{\tau}(-I + W).$$

Thus each eigenvalue  $\lambda$  of  $W$  corresponds to a Jacobian eigenvalue

$$\mu = \frac{\lambda - 1}{\tau}.$$

By Girko's circular law, the eigenvalues of  $W$  approximately fill the disk

$$|\lambda| \leq g.$$

Therefore, the Jacobian eigenvalues approximately fill the shifted and rescaled disk

$$|\tau\mu + 1| \leq g.$$

The rightmost point of this disk is

$$\mu_{\max} \approx \frac{g - 1}{\tau}.$$

If  $g < 1$ , all Jacobian eigenvalues have negative real part, so the internal activity decays back to  $x = 0$ . If  $g > 1$ , some directions become locally unstable, so small

# Interpretation with Oja-Type Plasticity

For Adi to-do (Basically, with fast-slow dynamics this is approximately justified)

# Heuristic: Oja Learning Tracks Activity Correlations

For fixed activity  $x$ , the Oja-type rule

$$\dot{W}_{ij} = \eta x_i (x_j - x_i W_{ij}) - \lambda W_{ij}$$

has equilibrium

$$W_{ij}^* = \frac{\eta x_i x_j}{\eta x_i^2 + \lambda}.$$

Equivalently,

$$W^* = u x^T, \quad u_i = \frac{\eta x_i}{\eta x_i^2 + \lambda}.$$

Thus, for constant activity, the learned matrix is rank at most one. This does not require the approximation  $\eta x_i^2 \ll \lambda$ .

For time-dependent driven activity,  $x(t)$ , the analogous heuristic is

$$W_{ij}^* \approx \frac{\eta \langle x_i(t) x_j(t) \rangle}{\eta \langle x_i(t)^2 \rangle + \lambda}.$$

Heuristically, Oja plasticity tends to align  $W(t)$  with the normalized correlation structure of the activity. Low-rank collapse is expected only if the driven activity is

# Driven Dynamics: Linearization

With time-dependent input, the dynamics are

$$\tau \dot{x} = -x + \tanh(Wx + b(t)).$$

Around a driven trajectory  $x^*(t)$ , the perturbation dynamics satisfy

$$\tau \delta \dot{x} = [-I + D(t)W] \delta x,$$

where

$$D(t) = \text{diag}(\tanh'(Wx^*(t) + b(t))) = \text{diag}(\text{sech}^2(Wx^*(t) + b(t))).$$

Thus the effective recurrent matrix is

$$D(t)W.$$

# Driven Dynamics: Effective Random-Matrix Gain

If  $W_{ij}$  has variance  $g^2/n$ , then TODO WHAT DOES CONDITIONALLY MEAN HERE conditionally on  $D(t)$ ,

$$\text{Var}((D(t)W)_{ij}) = \frac{g^2 d_i(t)^2}{n}, \quad d_i(t) = D_{ii}(t).$$

This suggests the heuristic effective gain

$$g_{\text{eff}}(t) \approx g \sqrt{\frac{1}{n} \sum_{i=1}^n d_i(t)^2}.$$

Girko's law gives  $g = 1$  as the undriven fixed-weight reference threshold. In the driven nonlinear system, saturation changes the effective gain through  $D(t)$ , so instability is better tracked by  $g_{\text{eff}}(t)$  than by  $g$  alone.

# Oja Learning Does Not Self-Organize to the Edge of Chaos

A key motivation for studying Oja-type plasticity in recurrent networks is the conjecture that local learning rules might spontaneously drive the network toward a critical operating point near the edge of chaos, where  $\rho(W) \approx 1$ .

To test this, we sweep initial spectral radii  $g_0 \in \{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 2.5\}$ , initialize  $W_{ij} \sim \mathcal{N}(0, g_0^2/n)$ , and run the full coupled system

$$\tau \dot{x}_i = -x_i + \tanh\left(\sum_j W_{ij} x_j + A \sin(\omega t + \phi_i)\right), \quad \dot{W}_{ij} = \eta x_i (x_j - x_i W_{ij}) - \lambda W_{ij},$$

with  $A = 1$  (driven) and  $A = 0$  (autonomous) for  $T = 400$  time units.



## Finding

Under sinusoidal drive,  $\rho(W) \rightarrow 45$  for every initial  $g_0$ . The drive erases all memory of the initial condition. The autonomous system ( $A = 0$ ) does exhibit a clean bifurcation at  $g_0 = 1$ , but the addition of

# Why the Drive Causes Runaway Growth

**Separation of timescales.** Activity relaxes on  $\tau = 1$ ; weights evolve on  $1/\eta = 100$ ; since  $\eta\tau = 0.01 \ll 1$ ,  $W$  is approximately constant on the fast (activity) timescale. By the averaging method we hold  $W$  fixed and study the fast  $x$ -subsystem. *Relaxes* means transients from initial conditions decay exponentially at rate  $(1 - \rho(DW))/\tau$ , leaving only the driven response. For  $g < 1$ ,  $\rho(DW) \leq \rho(W) < 1$  (since  $d_i \leq 1$ ), so the Poincaré map is a contraction and by the Banach fixed-point theorem there is a *unique* globally attracting periodic orbit at frequency  $\omega$ ; for  $g > 1$  the argument is heuristic. Averaging over one drive period  $T_{\text{avg}} = 2\pi/\omega$  gives the correlation matrix felt by the slow  $W$  dynamics:

$$C(W) = \frac{1}{T_{\text{avg}}} \int_0^{T_{\text{avg}}} x(t; W) x(t; W)^T dt.$$

The weight dynamics on the slow manifold reduce to

$$\dot{W}_{ij} \approx \eta [C_{ij}(W) - C_{ii}(W) W_{ij}] - \lambda W_{ij}.$$

Setting this to zero gives the fixed point

$$W_{ij}^* = \frac{\eta C_{ij}(W^*)}{\eta C_{ii}(W^*) + \lambda}.$$

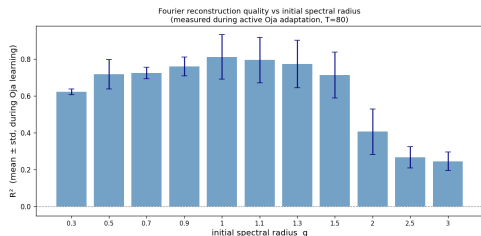
# Edge-of-Chaos Optimality for Signal Reconstruction

Although Oja learning does not maintain the network at the edge of chaos, the initial spectral radius  $g$  still determines reconstruction quality.

For each  $g$ , we run the driven Oja system for  $T = 80$  time units and fit a linear readout  $c^* \in \mathbb{R}^n$  to reconstruct the target signal  $y(t) = \sin(\omega t)$  from the activity history  $X \in \mathbb{R}^{T/dt \times n}$ :

$$c^* = \arg \min_c \|Xc - y\|^2, \quad R^2 = 1 - \frac{\|Xc^* - y\|^2}{\text{Var}(y)}.$$

The window  $T = 80$  is the natural measurement regime: transients have decayed (since  $\tau = 1$ ) but Oja has not yet fully saturated the network ( $\rho(W) \approx 10$ – $20$ ).



## Finding

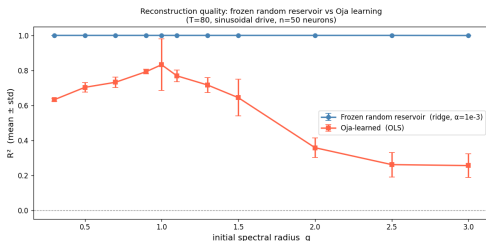
$R^2$  peaks at  $g \approx 0.9$ – $1.0$ , reaching  $R_{\max}^2 \approx 0.81 \pm 0.06$ , and falls to  $R^2 \approx 0.22$  at  $g = 3.0$ . Networks initialized at the edge of chaos produce the richest activity during the learning transient, enabling more



# Frozen Random Reservoir Outperforms Oja-Learned Weights

To assess the value of Oja learning directly, we compare to a frozen baseline. For each  $g$ , we run the same driven network with weights held fixed ( $\eta = 0$ ) and fit a ridge-regression readout with regularization  $\alpha = 10^{-3}$ :

$$c^* = (X^T X + \alpha I)^{-1} X^T y.$$



The frozen reservoir achieves  $R^2 \approx 1.0$  for every  $g$ . This is not a trivial result: even a static random matrix, when driven by  $A \sin(\omega t)$ , projects the input into a high-dimensional space from which the target frequency can always be recovered by a linear readout.

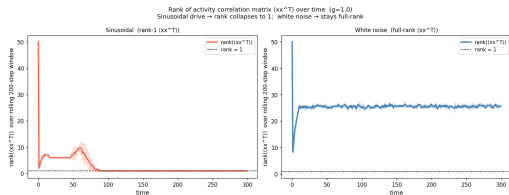
## Finding

Oja learning is counterproductive.  
It drives  $\rho(W) \rightarrow 45$ , saturating

# Input Structure Drives Rank Collapse: A Mechanistic Test

The slow-manifold analysis predicts that rank collapse is tied to the rank of  $C(W)$ , not to the learning rule per se. We test this by replacing the sinusoidal drive with three alternatives:

- *Quasi-periodic*:  $A/\sqrt{3} \sum_{k=1}^3 \sin(\omega_k t + \phi_i)$ , with  $\omega_1 = 1$ ,  $\omega_2 = \varphi$  (golden ratio),  $\omega_3 = \sqrt{2}$ .
- *White noise*:  $\xi_i(t) \sim \mathcal{N}(0, A^2/2)$  independently each step.
- *Zero drive*:  $A = 0$  (autonomous network).

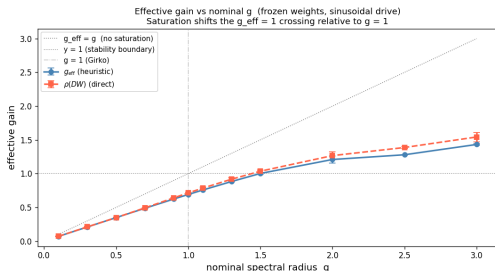


## Finding

For  $g < 1$  (stable regime), white noise keeps  $\text{rank}(\hat{C}) \approx n$  and prevents rank collapse entirely:  $\rho(W) \rightarrow 0$ ,  $\text{PR}(W) \approx 36$ . Sinusoidal drive collapses  $\text{rank}(\hat{C}) \rightarrow 1$  within  $T \approx 50$ , and  $\rho(W) \rightarrow 43$  in all cases. For  $g > 1.5$  (chaotic regime),

# Effective Gain Is Self-Regulated by Saturation

The heuristic  $g_{\text{eff}}(t) \approx (\|W(t)\|_F / \sqrt{n}) \sqrt{\text{mean}(d_i^2)}$  was tested in two ways: tracking it during Oja learning, and scanning frozen-weight networks across  $g$ .



During Oja learning with sinusoidal drive,  $\rho(W) \rightarrow 43$  but every neuron saturates ( $|h_i| \rightarrow \infty$ ), so  $d_i = \text{sech}^2(h_i) \rightarrow 0$  and  $g_{\text{eff}} \rightarrow 0$ . The network is in a *maximally saturated* fixed point, not in a chaotic state — the linearized effective dynamics are always stable despite the exploding nominal spectral radius.

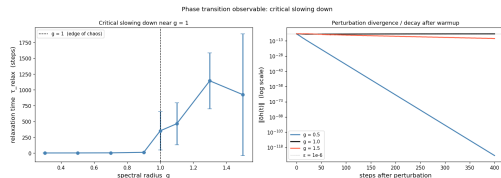
## Finding

During Oja learning,  $g_{\text{eff}} \rightarrow 0$  as the network saturates — the Oja fixed point is dynamically stable despite  $\rho(W) \rightarrow 43$ .

# Critical Slowing Down at the Phase Transition

Independent evidence for the  $g = 1$  phase transition comes from measuring the relaxation time of the autonomous discrete map  $h_{t+1} = Wf(h_t)$ .

A small perturbation  $\delta h(0)$  evolves as  $\|\delta h(t)\| \sim \|\delta h(0)\| e^{\lambda_1 t}$  in the linear regime, where  $\lambda_1$  is the largest Lyapunov exponent. For  $\lambda_1 < 0$  the relaxation time is  $\tau_{\text{relax}} = -1/\lambda_1$ ; for  $\lambda_1 > 0$  the perturbation grows and  $\tau_{\text{relax}}$  is undefined.



| $g$  | $\lambda_1$ | $\tau_{\text{relax}}$ |
|------|-------------|-----------------------|
| 0.30 | -1.20       | 0.8 steps             |
| 0.70 | -0.35       | 2.9 steps             |
| 0.90 | -0.10       | 10.8 steps            |
| 1.00 | -0.005      | $\sim 350$ steps      |
| 2.00 | +0.10       | (chaotic)             |

## Finding

$\tau_{\text{relax}}$  diverges as  $g \rightarrow 1^-$ , which is the defining signature of a continuous (second-order) phase transition. The network at  $g = 1$  has no preferred timescale for

# Summary

We have studied a recurrent network in which both activity and weights evolve simultaneously under Oja-type plasticity and sinusoidal drive. The main findings are:

- 1 Oja learning *does not* self-organize the spectral radius toward the edge of chaos. Under sinusoidal drive,  $\rho(W) \rightarrow 45$  regardless of the initial  $g_0$ , because the drive creates a persistent rank-one correlation structure that Oja amplifies without bound.
- 2 The initial spectral radius  $g$  still determines signal reconstruction quality:  $R^2$  peaks near  $g = 1$ , where network activity is richest during the early learning transient.
- 3 A frozen random reservoir with ridge regression outperforms the Oja-learned network for all  $g$ , because Oja learning destroys the activity diversity that makes reconstruction tractable.
- 4 Rank collapse requires *structured* periodic input in the stable regime ( $g < 1$ ): white noise drives do not collapse  $\rho(W)$  or reduce the participation ratio. In the chaotic regime ( $g > 1.5$ ), collapse occurs regardless of input structure due to the dominant Lyapunov mode.

# Open Questions

Several questions remain for future work.

The fixed-point analysis shows that  $W^* \approx \frac{\eta}{\lambda} C(W^*)$  on the slow manifold. A full stability analysis of this self-consistent equation would clarify whether the  $\rho(W) \rightarrow \infty$  trajectory is the only attractor under sinusoidal drive, or whether there exist bounded fixed points at large  $n$ .

The quasi-periodic drive collapses  $\rho(W)$  as strongly as the single sinusoid, which suggests the number of active modes in  $C$  is less important than whether any structured periodic components are present. A precise criterion — characterizing exactly which input power spectra lead to rank collapse — would sharpen the connection to the seizure hypothesis.

## Seizure connection

The key experimental prediction is: rhythmic, periodic input is necessary for Oja-driven rank collapse in stable networks, while random input is not sufficient. This suggests that pathological periodicity — rather than high firing rates alone — may be the mechanistically important feature of seizure-like activity driving network-level dimensionality collapse.